

1. Create a table listing 5 common network threats (such as phishing, DDoS, man-in-the-middle, malware, and password attacks), and for each, write one real-world example from an app or service you use (like WhatsApp, Instagram, or Paytm).

Network Threat	Description	Real-World Example
Phishing	Fake emails, websites, or messages that steal login credentials.	A fake Paytm SMS asks you to click a link and enter your account details.
DDoS (Distributed Denial of Service)	Attackers flood a server with massive traffic, making the service unavailable.	During a sale, attackers send huge traffic to BookMyShow , causing the website to become slow or unavailable.
Man-in-the-Middle (MITM)	An attacker secretly intercepts communication between two parties.	Using public Wi-Fi, an attacker tries to intercept your WhatsApp Web session if the network is insecure.
Malware	Malicious software that damages or steals data from a device.	Downloading a fake Instagram app from an unofficial website installs malware on your phone.
Password Attack	Attackers try to guess or steal passwords to gain unauthorized access.	Someone repeatedly tries different passwords to log into your Instagram account.

**2. Simulate a brute-force attack scenario by writing a simple Python script that tries to guess a 4-digit PIN code by checking all possible combinations against a preset value.

Constraint: The script should print the number of attempts it took to guess the correct PIN.**

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# Brute Force Attack Simulation
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correct_pin = "4827"
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```
attempts = 0
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```

for i in range(10000):
    guess = str(i).zfill(4) # Convert number to 4-digit format
    attempts += 1

    if guess == correct_pin:
        print("PIN Found:", guess)
        print("Number of Attempts:", attempts)
        break

```

Sample Output

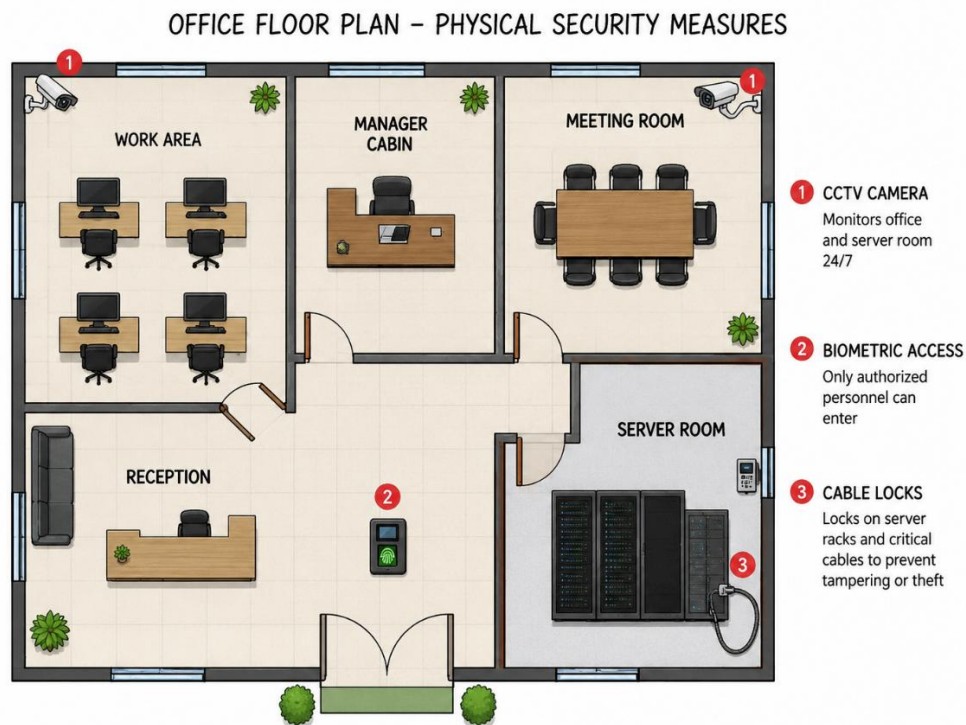
PIN Found: 4827

Number of Attempts: 4828

3. List 4 physical security risks that could affect a server room hosting a popular app like Zomato or BookMyShow, and suggest one mitigation step for each risk.

Physical Security Risk	Example (Zomato/BookMyShow Server Room)	Mitigation
Fire	Electrical short circuit damages servers.	Install automatic fire detection and suppression systems.
Unauthorized Access	An intruder enters the server room and steals hardware.	Use biometric access control and CCTV surveillance.
Power Failure	Servers shut down due to electricity outage.	Install UPS and backup generators.
Flood or Water Leakage	Water damages servers and networking equipment.	Keep servers on raised racks and install water leak detection sensors.

4. Draw a simple floor plan (hand-drawn or digital) of a small office and mark at least 3 physical security measures you would implement to protect network equipment (such as CCTV, biometric access, or cable locks).



5. Use ChatGPT or a similar AI tool to generate a checklist of 7 actions a network admin should take to mitigate risks from both network threats and physical security breaches, then review and edit the checklist to make it specific for a college computer lab environment.

No.	Action
1	Use strong passwords and enable multi-factor authentication (MFA) for all administrator accounts.
2	Install and regularly update antivirus software on every computer.
3	Keep Windows, Linux, and all software updated with the latest security patches.
4	Restrict access to the server room using biometric locks or authorized ID cards only.

5	install CCTV cameras to monitor the computer lab and server room.
6	Perform regular backups of student data, lab configurations, and important files.
7	Educate students and staff about phishing emails, safe internet usage, and reporting suspicious activities.